The AI Operating System

How one person runs a full digital agency — websites, video, ads, automation, outreach — on top of Claude Code. The daily stack, the rules, and the workflows, written for a friend who wants to copy it.

PRACTITIONER GUIDE · V1.0 · AUGUST 2026

> This isn't "10 AI tools you should try." It's the actual operating system: how work gets decomposed, which agent does it, what gate it must pass before it goes out, and the hard-won rules that stop expensive mistakes from repeating. Everything here is generic — no client names, no credentials, no private data.

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1 — The Core Idea

Most people use AI as a **chat box**: ask, get text, paste somewhere, forget. That caps out fast, because the model starts every conversation with amnesia and no hands.

The shift is to treat AI as an **operating system for the business**. Three things change:

It remembers

Durable facts live in files on disk, not in the chat. Every new session loads the index automatically.

It has hands

Terminal, filesystem, git, browser, ffmpeg, APIs. It doesn't describe the fix — it makes it and verifies it.

It has procedures

Repeated workflows are frozen into skills. You stop re-explaining and start invoking.

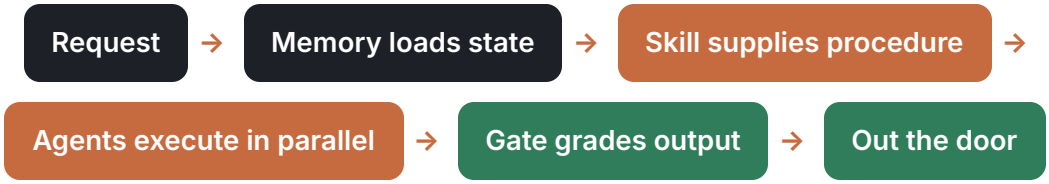
> **The governing principle**
>
> A prompt you have typed three times is a bug. Convert it into a skill (a procedure) or a memory (a fact). The system should get cheaper to operate every week, not more expensive.

What "one-person agency" actually means here

Website builds, video editing, paid ads, cold outreach, client reporting, and internal ops all run through one terminal. The human does three things the machine can't: pick the direction, approve money and outward-facing sends, and make taste calls. Everything between those decisions is delegated.

2 — The Four Layers

LAYER	WHAT IT IS	WHY IT EXISTS
Memory	Markdown files on disk + a thin index loaded every session	So the assistant knows the state of the business without being re-briefed
Rules	A global instruction file — facts that are *always* true	Non-negotiables: how to behave, what never to do, house standards
Skills	Versioned procedure files, invoked as `/name`	Repeatable multi-step workflows with their gotchas already baked in
Agents	Specialist sub-assistants with their own context + tool set	Parallelism and separation of concerns — a builder never grades itself



The critical distinction

Rules file = facts always true. "Never freeze the machine during a render." "Read code before modifying it."
Skill = a procedure you'd otherwise re-paste. "Here are the 14 steps to turn a raw phone clip into a finished vertical video with captions."

Getting this wrong is the most common failure. People dump procedures into the always-loaded rules file, it balloons to thousands of tokens, and every single session pays for instructions it doesn't need.

3 — Memory: The Brain

Memory is a folder of small markdown files. One fact per file. Each file has a name, a one-line description, and a type. A single `MEMORY.md` index — one line per memory — is the only thing loaded at session start.

Four memory types

TYPE	CONTENTS	EXAMPLE SHAPE
user	Who the operator is: role, expertise, standing preferences	"Prefers deliverables as self-contained HTML, not markdown"
feedback	Corrections and confirmed approaches — *always with the why*	"A commit proves nothing about production. Check the live environment."
project	Ongoing work, goals, constraints not derivable from the code	"Build X is blocked pending a design decision from the client"
reference	Pointers: dashboards, tickets, docs, environment facts	"The render library lives at `_lib/` and exposes these helpers"

The rules that keep memory useful

- **One fact per file.** Bundled files never get updated correctly and never get deleted.
- **Never store what the repo already records.** Code structure, past fixes, git history — the machine can read those. Memory is for what *isn't* written down.
- **Convert relative dates to absolute.** "Next week" is meaningless three months later.
- **Link liberally** with wiki-style links. A link to a memory that doesn't exist yet is a to-do, not an error.
- **Delete wrong memories immediately.** A stale fact is worse than a missing one, because it's trusted.

> **The highest-leverage memory type is *feedback***
>
> Every time the system makes an expensive mistake, the fix is not "be more careful next time" — it's a feedback memory with the trigger, the failure mode, and the why. That mistake then structurally cannot recur. This is the compounding mechanism of the entire system.

Checkpoints

At the end of a working session, state gets written to a dated checkpoint file: what got done, what's blocked, what's next. Tomorrow's session opens with yesterday's ending. No re-briefing, no "where were we."

4 — Skills: Frozen Procedures

A skill is a markdown file with frontmatter (name, description, trigger phrases) and a body (the steps). When a request matches the description, it loads automatically. You can also invoke it directly with `/name`.

Anatomy of a good skill

```
---
name: finish-video
description: Turn a finished cut into per-platform outputs.
  Trigger on "finish this video", "wrap the reel", "/finish-video",
  or when a final master exists and needs distribution assets.
---

## Preconditions
- Master file exists and passed the render gate
- Captions verified against the RENDERED file, not the source

## Steps
1. Probe the master (codec, resolution, audio rate) and ASSERT, don't print
2. Run the copyright gate on the audio bed
3. Generate per-platform captions from the copy template
4. Build the schedule payload
5. Hand to the human for the final send

## Gotchas
- Audio normalization can silently resample; force the output rate
- A "success" exit code from the encoder is not proof of a valid file
```

What separates a skill that fires from one that never does

Strong triggers

The description must contain the literal phrases a human would actually type. Vague descriptions never auto-invoke, and an uninvoked skill is dead weight.

Gotchas section

This is the entire value. Anyone can write the happy path. The skill earns its keep by carrying the six things that went wrong the first four times.

Runnable steps

Real commands with real paths, not "then process the file." If a step needs a human decision, say so explicitly.

Preconditions

State what must be true before step 1. Half of all skill failures are a missing input nobody checked for.

Categories in daily use

CATEGORY	WHAT IT COVERS
Build	Site scaffolds, design systems, deploy chains, client onboarding, repo QA
Video	Cut, caption, grade, overlay, thumbnail, copyright check, ship gate, distribution
Content	Post packs, carousels, long-form scripts, humanizer pass, scheduling
Ads	Plan, launch, monitor, fatigue detection, budget reallocation, kill rules
Ops	Daily brief, realign/prioritize, machine audit, storage audit, checkpoint
Meta	Skill-builder (build new skills), critic-loop (grade anything), council (multi-model)

The count matters less than the discipline: every skill exists because a workflow was performed by hand at least twice and hurt the second time.

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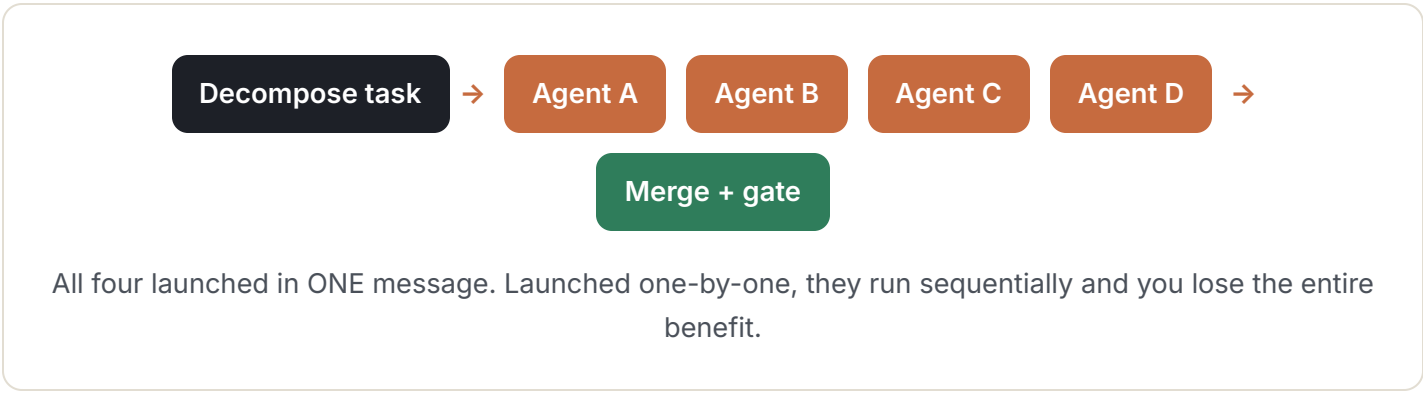
5 — Subagents & Parallelism

A subagent is a separate assistant instance with its own context window, its own tool permissions, and a narrow mission. It reports back a synthesis — not a dump of everything it read.

Why this is the single biggest speed unlock

Sequential work is the default and it's almost always wrong. If four sub-tasks have no dependencies between them, they should run at the same time. Four 30-minute tasks become 30 minutes of wall-clock,

not two hours.



All four launched in ONE message. Launched one-by-one, they run sequentially and you lose the entire benefit.

Elite briefs, not baby prompts

The quality of a subagent's output is almost entirely determined by the brief. A thin brief ("look into the auth thing") returns thin work. A full brief has six parts:

1. **Mission + done-definition** — what "finished" concretely looks like
2. **Full context handoff** — the agent can't see your conversation; give it everything
3. **Autonomy mandate** — explicitly authorize it to make decisions without checking back
4. **Quality bar** — what standard it's being held to
5. **Right specialist** — a security auditor, not a generalist, for a security sweep
6. **Return contract** — "return a synthesis, decisions made, and artifact paths — not file contents"

Specialist roster (generic pattern)

AGENT	MISSION
Researcher	Multi-source web research in an isolated context, returns synthesis only. Keeps 40 pages of scraped text out of the main conversation.
Explorer	Read-only fan-out search across a large codebase. Locates things; doesn't judge them.
Builder specialists	Framework-specific implementers — a WordPress builder, a frontend builder, a TypeScript specialist.
Auditors	Security sweep, site pre-ship audit, content/copy QA. All read-only by design.
Planner / executive	Decomposes a fuzzy goal into a routed, prioritized execution plan. Plans only — never builds, never spawns.
Curator	Keeps the memory vault clean: orphans, stale entries, broken links.

> **An agent never grades its own work**
>
> The builder is invested in the build. Grading always goes to a fresh-context agent that didn't write the thing, ideally one instructed to default to FAIL on uncertainty. This single rule catches more defects than any other practice in the system.

Worktree isolation

When multiple agents edit files in parallel, they collide. The fix is git worktrees: each agent gets its own isolated checkout, does its work, opens its own change. Merge smallest-first so later rebases stay trivial. Blast radius per agent is one worktree; a crashed agent doesn't take down the other three.

6 — The Daily Loop



Morning: context is already loaded

A session-start hook injects today's priorities, the daily note, and any follow-ups that have gone silent past their cadence. There is no "let me catch you up" phase. The first message of the day is already about work.

The follow-up re-fire

The most valuable automation in the entire stack is embarrassingly simple: a check that lists every outstanding item with no reply past its follow-up window, and drafts the nudge. Deals die from unfired follow-ups far more often than from rejection. Automating the *reminder to follow up* — not the sending — beats automating almost anything else.

Evening: realign

A nightly job reads the memory index, the day's activity, and rewrites the entire task universe into one ranked plan for tomorrow: revenue first, blocked-on-human items surfaced explicitly, dead items culled, stale entries flagged. That plan is what the morning hook injects. The loop closes.

Working modes

Plan mode gate

Before any billable or multi-file change: draft the plan, get it approved, *then* execute. The plan doubles as the client sign-off artifact and the billing record.

Compressed mode

A terse output mode for long sessions. Drops filler and hedging, keeps all technical substance. Meaningful token savings on multi-hour work.

Context hygiene

Clear when switching projects. Compact at milestones. A context stuffed with three unrelated projects produces confused work.

Capture-before-close

Any session that produced a hard-won fix writes it to a skill or memory *before* the session ends. Otherwise it evaporates and gets re-learned at full price.

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7 — Web & Client Delivery



Onboarding is a hard gate

Every new client repo starts with a structured onboarding pass that captures the thing everyone forgets until it breaks: **the deploy target and its specific quirks**. Some hosts take a git push. Some take a manual archive upload. Some need a cache purge afterward or the old build stays live. Writing that down once, per client, at the start, prevents the "why is my fix not showing" hour later.

Speculative builds beat proposals

For a competitive pitch, a working demo outperforms a document. But two rules are absolute:

> **Never let a spec demo tell a lie**
>
> If a demo shows metrics, testimonials, or logos, they must be labelled as illustrative above the fold — not buried in a footer. Fabricated proof in a pitch is the fastest way to lose a client you were about to win, and it can outlive the pitch on the open web.

Pre-ship audit dimensions

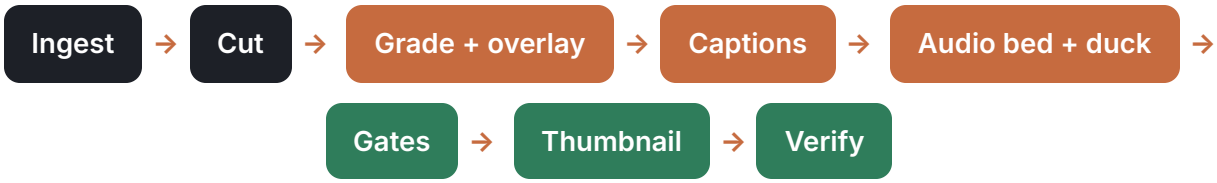
DIMENSION	CHECKED FOR
Responsive	Real breakpoints on real viewports — not a desktop window resized
Links	Every internal and external link resolved; no 404s, no placeholder anchors
SEO / structure	Meta, heading hierarchy, structured data, canonical, sitemap
Performance	Core Web Vitals signals, image weight, render-blocking assets
Leftovers	Lorem ipsum, TODO comments, dummy emails, test data
Accessibility	Contrast, focus states, alt text, keyboard reachability
Security	Exposed keys, dependency advisories, misconfigured headers

Deliverable format

Client-facing documents ship as **self-contained HTML with a print-to-PDF button**, on a bright theme. Not markdown, not a raw doc. One file, zero dependencies, opens anywhere, prints clean. A client should never need a tool they don't have to read your work. (This guide is built exactly that way.)

8 — Video Production Pipeline

Video is where the most scar tissue lives, because the failure modes are silent. A render can "succeed," produce a file, and still be unusable — wrong audio rate, missing frames, a privacy leak in one corner of one shot.



The gates that actually catch things

GATE	WHAT IT ASSERTS
Pixel gate	A frame-by-frame pass over the *rendered* file at a real sample rate. Re-reading the audio proves nothing about the picture.
Audio spec gate	Asserts sample rate, channel layout, and a loudness floor on the final file. Normalization filters can silently change the output rate.
Copyright gate	Fingerprints the music bed *before* upload. Catching a claim beforehand costs minutes; catching it afterward costs the post.
Privacy gate	Screens, whiteboards, documents, faces, filenames. Re-run after *any* reframe.
Publication check	Confirms the live post matches the intended master by frame hash — not by duration.

> **Assert, don't print**
> A gate that prints a property and lets a human eyeball it is not a gate. It must fail loudly on mismatch. "It printed 48000" and "it asserted the rate is 48000" are different programs, and only one of them catches the bug at 2am.

Clearance is resolution-specific

If you clear a video at one framing and later widen the crop or re-derive at higher resolution, every clearance is void. The wider frame contains pixels nobody has ever checked. Re-derive geometry from the original source and re-run every gate.

Hybrid rendering

The pattern that works: assemble the heavy video with a fast native encoder, render the fancy animated overlay separately as a transparent alpha layer, then composite. Pushing 4K source through a JavaScript animation renderer chokes; keeping them in separate lanes gives you both premium motion and a render that finishes.

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9 — Content & Distribution

The engine

One command produces a full day's pack across every channel — long post, carousel, short-form script, platform-native variants. Voice is locked to a template so output stays recognizably human and consistent rather than drifting into generic assistant prose.

Three passes every piece must survive

Humanizer

Strips AI tells: the em-dash tic, "in today's fast-paced world," tri-colon structures, hollow transitions, over-hedged claims.

No-fake-claims

Every number, result, and testimonial traces to something real. Unverifiable claims get cut, not softened.

Anti-repetition

Cross-day check so the same hook, anchor, or anecdote doesn't recycle within a window. Readers notice before you do.

Specificity is the authenticity signal

Generic content reads as machine-written regardless of how well it's phrased. Concrete detail woven into the body — a real place, a real number, a real constraint you hit — is what makes it land. Woven, not

pasted: a specific sentence dropped into an otherwise generic paragraph reads worse than no specificity at all.

Distribution rules learned the hard way

- **Platform classifiers move.** A CTA pattern that worked for a year can start suppressing reach overnight. Retire tactics on evidence, not on sentiment.
- **Scheduling clock is the account's timezone, not yours.** Read it live from the API every build. Assuming it burns an entire drip.
- **Test row one before trusting the batch.** A ten-post schedule built on a column misalignment produces ten broken posts.
- **Publishing must be idempotent.** If a job re-runs, it must not re-fire. Duplicate posts are worse than late posts.

10 — Automation (n8n / VPS)

Long-running automations — inbox triage, lead pipelines, scheduled sweeps — live on a small VPS running a workflow engine, not on the laptop. The laptop is for building; the server is for running.

The loop gate — three caps and a breaker

> **No automation loop goes live without all four**
>
> 1. **Step cap** — maximum iterations per run
> 2. **Wall-clock cap** — maximum runtime before forced stop
> 3. **Spend cap** — maximum cost per run and per day
> 4. **Circuit breaker** — an error-rate or anomaly threshold that halts the loop and alerts a human
>
> All four live in **config**, never in a prompt. A cap that exists only as an instruction in a prompt is not a cap — it's a suggestion to a system that can misread it.

Monitor distribution, not liveness

"The workflow ran successfully" is a near-useless signal. A workflow that ran 40 times and produced identical empty output every time reports as perfectly healthy. Monitor the *shape of the output*: how many items, what variety, what changed versus yesterday. Silent degradation is the normal failure mode of automation, and liveness checks are blind to it.

Outbound discipline

- **Low daily volume, warmed sender.** Volume is what kills domains, not content.
- **Validate addresses before sending.** Bounce rate is the reputation metric that matters.
- **Log every send to a ledger.** Without it you cannot answer "did we already contact them" and you will double-send.
- **Poll every folder, not just the inbox.** Replies land in spam and get missed for weeks.

- **AI review is a sensor, never a verdict.** It flags, scores, and drafts. A human presses send on anything that leaves the building.

> **Audit the running estate, not the plan file**
> The plan says a workflow is active. The plan is a document; it has no idea what the server is doing. Before reporting status, query the actual engine, the actual mailbox, the actual DNS. Documents drift from reality within days, and a status report built from a document is fiction.

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11 — Paid Ads Discipline

Tracking before spending

The most expensive ad mistake possible: run a campaign for days, get clicks, get zero recorded conversions — and discover the conversion event never fired because the tracking code was written but never deployed, or an environment variable was never set in production.

> **Fire the conversion event manually and see it land before spending a cent**
> Not "the code is committed." Not "it looks right." Trigger the real event, watch it arrive in the platform's event dashboard, then turn on budget.

Rules set before launch, not during

RULE	SET IN ADVANCE
Kill threshold	Spend with zero conversions at which an ad is paused — decided before launch, executed without renegotiation
Learning floor	Minimum data before *any* optimization decision. Reacting to day-one numbers is reacting to noise.
Fatigue signal	Frequency and CTR-decay thresholds that trigger a creative refresh
Reallocation cadence	How often budget moves and by what maximum step
Max concurrent tests	A hard cap. More variants than you have traffic for teaches you nothing, slowly.

Read-only by default

The ads integration is configured read-only. Reporting, analysis, and recommendations are automated; changing budgets or activating campaigns requires a human. An automation bug that misreads a metric costs a wrong report. The same bug with write access costs real money while you sleep.

12 — Quality Gates

Nothing billable, deployable, or public goes out without an independent grade from something that did not build it.

The critic loop



Loops until pass or a round limit. Judges default to FAIL on uncertainty.

Design principles behind the gate

- **A false pass costs more than a false block.** Calibrate the whole gate around that asymmetry.
- **Multiple judges, reasons before scores.** Forcing the reasoning first prevents a number being picked and then justified.
- **A verification unit kills hallucinated faults.** Critics invent problems too. Every finding must be reproducible against the artifact or it's discarded.
- **Resolved findings are tracked.** Otherwise round three re-raises what round one fixed and the loop never converges.
- **Auto-fail overrides every vote.** A verified exposed secret or an unconsented identifiable face is a hard stop regardless of how good everything else is.

Green is not the same as good

> **Passing gates measures the absence of defects, not the presence of quality**
>
> A video can clear every technical check — correct codec, clean audio, no privacy leak — and still be badly edited. A site can pass every audit and still be ugly and unconvincing. Automated gates catch *failures*. Craft still requires a human with taste to watch it, read it, and say yes. Never mistake a green board for a good deliverable.

Multi-model review

For high-stakes work, a second and third model from different providers review the same artifact. Different training produces genuinely different blind spots; agreement across families is a much stronger signal than one model's confidence. Used selectively — it costs real tokens and most work doesn't need it.

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13 — Hard Rules (Scar Tissue)

Each of these exists because it was learned expensively. This section is the most portable part of the document — these apply to almost anyone doing AI-assisted work.

Committed is not deployed

A commit proves nothing about production. Before writing a "missing" feature, check for unpushed commits and check whether the environment variable the code depends on is actually set in the live environment. Fail-soft code produces silence, not errors — so a broken integration looks exactly like a working one that had nothing to do.

Verify against reality, not against your notes

Query the running system, the live mailbox, the actual DNS record. Never repoint an endpoint without fetching what's currently there first — the "parking page" you're about to overwrite may be the live site.

Sent is not pending

Once something is sent, it's awaiting a reply — not awaiting action. Tracking it as pending produces duplicate follow-ups that read as desperate. Status comes from the mailbox, not from a plan file.

A spot check is not proof

Verifying three files out of two hundred tells you three files are fine. If a claim covers a set, verify the set or state the sample size explicitly.

Regex sweeps hit more than intended

Bulk find-and-replace across a repo always catches something you didn't picture. Dry-run first, read the full match list, then apply.

Opening a file for writing can destroy it before the write fails

Truncating write modes empty the target the moment the handle opens. If the script then throws, the data is gone. Write to a temp file and atomically replace.

Recursive scans can abort silently on a permission error

Some tools return a clean zero-result on a partial failure. Zero results is a suspicious answer, not a reassuring one — verify the scan actually traversed everything before concluding "nothing found."

Filenames re-identify anonymized material

You blurred the faces and bleeped the names, then shipped a file whose name contains the company. Grep the *rendered output and its metadata*, not the source project.

Exit code zero is not success

Encoders, uploaders, and API clients all have paths that exit clean while producing garbage. Check the artifact, not the return code.

Local-only is not backed up

Know exactly which folders sync and which don't. The assumption that "it's in the cloud folder" is how irreplaceable footage disappears. Verify sync status per folder, in writing, once.

14 — Machine & Cost Discipline

Never freeze the box

> **Every heavy job reserves a fixed share of CPU for the human**
> Renders, encodes, batch processing — all run at lowered priority with a headroom check before starting and a worker pool capped below the core count. Never two heavy jobs at once.
>
> A render that takes 40 minutes instead of 25 is fine. A frozen machine for 25 minutes is a **failure**, because the human can't work, can't review, can't respond to a client. Throughput of the machine is not the metric. Throughput of the operator is.

Token discipline

- **Keep the always-loaded layer thin.** Every token in the global rules file is paid on every single session forever. Procedures belong in skills, loaded on demand.
- **Compressed output mode** for long sessions — meaningful savings, zero loss of substance.
- **Isolate research in subagents.** Forty pages of scraped text should never enter the main context; only the synthesis should.
- **Return contracts.** Agents return conclusions and file paths, never file dumps.
- **Don't re-read.** If a file was read this session and hasn't changed, reading it again is pure waste.
- **Match model to task.** Mechanical work doesn't need the expensive tier; the hardest reasoning and the final judge do.

Security hygiene

- Secrets in environment variables and secret stores — never in prompts, never in committed files, never in a plan document.
- Secret scanning in the pre-ship gate, on the *rendered output* as well as the source.
- Least-privilege tokens. Read-only wherever write isn't strictly required.
- A scheduled audit of the machine and the estate — what's exposed, what's stale, what's unpatched.
- Cookie-based access to logged-in services stays read-only. Read and draft; never auto-act on someone else's account.

15 — Build Your Own in 7 Days

Don't copy the whole thing. It took a long time and most of it is specific to one business. Build the spine, then let your own scar tissue fill it in.

DAY	BUILD	WHY THIS ORDER
1	Global rules file — facts always true, 20 lines max	Instant behavioral consistency. Resist the urge to make it long; you'll pay for every line forever.
2	Memory folder + index. Write 5 facts your assistant re-asks about	Kills the re-briefing tax immediately. Most visible day-one win.
3	Your first skill — the workflow you've explained most often	Teaches the format on a case where you already know every gotcha.
4	One quality gate on your highest-stakes output	An independent grader on the thing that costs most when it's wrong.
5	Parallel agents on one genuinely independent task set	Feeling four-at-once is what changes how you scope everything after.
6	A session-start hook injecting today's priorities	Closes the loop between sessions. This is what makes it feel like a system.
7	Checkpoint + realign routine	State survives the session. Compounding starts here.

The three habits that matter more than any tool

Write the failure down

Every expensive mistake becomes a memory with the trigger and the why — the same day. Not "be careful next time." A rule, in a file.

Never grade your own work

Independent review on anything that leaves the building. The builder cannot see the build clearly.

Parallel by default

Ask "do these depend on each other?" before every multi-part task. Usually they don't, and sequential was a habit not a requirement.

The one-line version

Give the AI memory so it stops forgetting, procedures so it stops improvising, parallelism so it stops queueing, and an independent judge so it stops shipping its own mistakes. Everything else is detail.

The AI Operating System — a practitioner's guide to running real work on Claude Code.
Generic by design: no client data, no credentials, no proprietary internals. Every rule here was learned by breaking something first.

Built as a single self-contained HTML file. Print to PDF from any browser.